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ITAI-2277

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Smarter game night predictions. NBA Player Performance Prediction Engine



Project Overview

Project Title:

CourtIQ – NBA Player Performance Prediction Engine

Project Type:

Student-Proposed Project (Option B)

Project Goal:

Court IQ is an NBA player performance prediction engine designed to analyze large-scale historical game data to forecast player outcomes on a given game night.

The system focuses on modeling minutes played as the primary driver of performance using machine learning techniques.



Problem Statement

Problem:

There is no explainable, minutes-first prediction system that models player performance using structured ML techniques.

Most NBA prediction methods:

- Rely on simple rolling averages
- Ignore rotation volatility
- Underestimate the importance of minutes

Proposed Solution

CourtIQ will:

- Predict expected minutes played
- Predict per-minute production
- Combine both to estimate:
 - Points (PTS)
 - Points + Rebounds + Assists (PRA)
- Provide over/under probability estimates
- Generate a confidence score

Core Formula:

Expected Stat = Predicted Minutes ×
Predicted Per-Minute Production

Technical Approach

Architecture

Data → Feature Engineering → Minutes Model → Production Model → Final Prediction

Models:

- XGBoost Regression
- Classification model (Over/Under probability)

Features:

- Rolling average minutes (5 & 10 games)
- Minutes volatility (std deviation)
- Usage trends
- Rest days / back-to-back
- Home vs Away
- Trained on multi-season NBA data (2000–present)

Data & Tools

Primary Data:

- NBA player game logs (all available seasons since 2000)
- Team-level statistics
- Game schedule and matchup context
- Thousands of player-game observations across multiple seasons

Tools:

- Python
- Google Colab (collaborative large-data processing)
- GitHub (version control)
- Scikit-learn / XGBoost

Timeline & Milestones

Phase 01 – Proposal (Week 3)

Project scope, technical plan, approval

Phase 02 – Data & EDA (Weeks 4–6)

Data pipeline, feature engineering, baseline analysis

Phase 03 – Working Model (Weeks 7–10)

Train minutes model + performance model

Phase 04 – Integrated Prototype (Weeks 11–13)

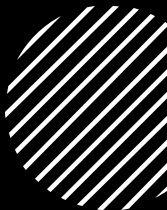
Full prediction engine + interface

Phase 05 – Final Delivery (Weeks 14–16)

Model evaluation, documentation, final presentation



Success Metrics



CourtIQ will be evaluated using:



Minutes prediction MAE (< 3 minutes)



Player performance MAE (PTS / PRA)



Over/Under classification accuracy



Improvement over rolling-average baseline



Model stability across multiple seasons

Risks & Ethical Considerations

Risks:

- Data inconsistency
- Overfitting
- Model generalization limits

Mitigation:

- Cross-validation
- Regularization
- Clear probability-based outputs

Ethics:

- Transparent uncertainty
- Not presenting predictions as guarantees
- Responsible AI use



Conclusion

CourtIQ will leverage all available NBA seasons since 2000, creating a large-scale dataset of thousands of player-game observations.

By applying:

- Multi-season historical data
- Structured feature engineering
- Minutes-first predictive modeling
- Cross-validation and benchmark testing

The system transforms high-volume NBA data into explainable, data-driven game night predictions. CourtIQ demonstrates scalable data processing, machine learning implementation, and full-system integration.